

went on to become very popular. Some of the most widely used distributions of Linux are Debian, Fedora, Linux Mint, Gentoo, PCLinuxOS, etc. Interestingly, MacOS and Linux are both based on the same software, which is Unix. Hence, users switching between the two make a smooth transition.

A little about Ubuntu

Developed by a company called Canonical, Ubuntu is based on one of the many derivatives of Linux-Debian. This software hit the market in 2004. Programmers understood the highlights of the OS, given the ease of programming that it provided. The command-line interface allowed them to use minimum syntax to write programs. Ubuntu has three versions, all of which were developed for different purposes.

- The desktop version of the OS was for the common masses.
- The server version was built for companies to maintain their databases.
- There was one more version that was created specifically for the programmers, and this was the core version of Ubuntu. This one was designed to support programmers in building robots and devices related to IoT.

Ubuntu and Linux

One mistake that people often make about Ubuntu and Linux is to confuse their nature. Users think that Linux is a distribution or an application that can be used independently on a computer. This is far from the truth. Linux is the kernel that is used to do the work within the distribution. A kernel is the innermost part of an operating system. This is the portion that communicates with the hardware to execute instructions given by a user. Enveloping the kernel is the shell, and the top-most layer is the application layer which consists of all the applications that one uses on their computer system. Ubuntu, on the other hand, is a distribution that runs on a Linux kernel. In simpler terms, Ubuntu is the entire operating system that incorporates all the three layers mentioned above.

Ubuntu v/s Windows

Unix and Microsoft have been competing to get the attention of users since their inception. With the advent of Microsoft's Windows, Unix was almost blown off due to several reasons. Windows had one significant advantage over Unix, or for that matter, all other Unix or Linux based distributions including Ubuntu, and this advantage was the ease of use. Let's look at some key differences.

- **Cost:** While both the operating systems are used widely, Ubuntu has an advantage when it comes to the price. Ubuntu is available for free for all the